EXPERIMENT 6: APPLY K-MEANS AND HIERARCHICAL CLUSTERING

1. DATASET SOURCE

The dataset used for this experiment is obtained from the Spotify dataset used for clustering analysis.

- **Source**: Spotify Audio Features Dataset (commonly available on platforms like Kaggle)
- It contains musical attributes of songs such as tempo, energy, loudness, danceability, etc.

2. DATASET DESCRIPTION

The dataset consists of multiple songs with their corresponding **audio features**, which are numerical representations of musical properties.

Key Features:

- **Danceability**: Measures how suitable a track is for dancing (0–1)
- **Energy**: Intensity and activity level of a song
- **Loudness**: Overall loudness in decibels (dB)
- **Speechiness**: Presence of spoken words
- **Acousticness**: Confidence measure of whether the track is acoustic
- **Instrumentalness**: Predicts if a track contains no vocals
- **Liveness**: Detects presence of an audience
- **Valence**: Musical positivity (happy vs sad)
- **Tempo**: Speed of the song (BPM)

Target Variable

- There is **no target variable** because clustering is an **unsupervised learning task**.

Dataset Characteristics

- Type: Numerical dataset
- Nature: Unlabeled
- Size: Depends on dataset version (typically thousands of songs)
- Use Case: Discover hidden patterns/groupings in music

3. MATHEMATICAL FORMULATION OF THE ALGORITHMS

K-Means Clustering

K-Means partitions data into **K clusters** by minimizing intra-cluster variance.

Objective Function:

$$J = \sum_{i=1}^{K} \sum_{x \in C_i} \| x - \mu_i \|^2$$

Where:

- C_i= Cluster i
- μ_i= Centroid of cluster i
- x= Data point

Algorithm Steps:

1. Initialize K centroids randomly
2. Assign each point to nearest centroid
3. Update centroids as mean of assigned points
4. Repeat until convergence

Hierarchical Clustering

Hierarchical clustering builds a tree-like structure called a **dendrogram**.

Distance Metrics:

- Euclidean Distance:

$$d(x, y) = \sqrt{\sum(x_i - y_i)^2}$$

Linkage Methods:

- **Single linkage**: Minimum distance
- **Complete linkage**: Maximum distance
- **Average linkage**: Mean distance

Process:

1. Treat each point as a cluster
2. Merge closest clusters iteratively
3. Continue until one cluster remains

4. ALGORITHM LIMITATIONS

K-Means Limitations

- Requires predefined number of clusters (K)
- Sensitive to initialization (random centroids)
- Assumes spherical clusters
- Poor performance with:
    - Non-linear data
    - Varying density clusters
    - Outliers

Hierarchical Clustering Limitations

- Computationally expensive ($O(n^2)$)
- Not scalable for large datasets
- Sensitive to noise and outliers
- Cannot undo merging (greedy approach)

5. METHODOLOGY / WORKFLOW

Step-by-Step Workflow

1. **Data Loading**
   - Import dataset into Python environment
2. **Data Preprocessing**
   - Handle missing values (if any)
   - Normalize/standardize features (important for distance-based methods)
3. **Feature Selection**
   - Select relevant numerical audio features
4. **K-Means Clustering**
   - Use Elbow Method to find optimal K
   - Train K-Means model
   - Assign cluster labels
5. **Hierarchical Clustering**
   - Compute distance matrix
   - Generate dendrogram
   - Choose number of clusters
   - Assign clusters
6. **Visualization**
   - Use PCA or 2D plots for visualization
7. **Evaluation**
   - Compare clustering results

6. PERFORMANCE ANALYSIS

Since clustering is unsupervised, evaluation is based on **internal metrics**:

1. Inertia (K-Means)

- Measures within-cluster sum of squares
- Lower value → better clustering

2. Silhouette Score

$$S = \frac{b - a}{\max(a, b)}$$

Where:

- a: Intra-cluster distance
- b: Nearest cluster distance
- Range: [-1, 1]
- Higher value → better clustering

Interpretation

- Well-separated clusters indicate meaningful grouping of songs
- Clusters may represent:
  - High-energy songs
  - Acoustic/soft songs
  - Dance tracks

7. HYPERPARAMETER TUNING

K-Means Hyperparameters

- **n_clusters (K)**:
  - Tuned using **Elbow Method**
- **init**:
  - k-means++ improves convergence
- **max_iter**:
  - Number of iterations
- **n_init**:
  - Number of centroid initializations

Tuning Impact

- Correct K improves clustering quality significantly
- Poor K leads to:
  - Underfitting (too few clusters)
  - Overfitting (too many clusters)

Hierarchical Clustering Hyperparameters

- **Linkage method** (single, complete, average)
- **Distance metric**

Tuning Impact

- Different linkage methods produce different cluster shapes
- Dendrogram helps visualize optimal cluster cut

CONCLUSION

This experiment demonstrates how **unsupervised learning techniques** like K-Means and Hierarchical Clustering can uncover hidden patterns in music data. K-Means is efficient and scalable, while Hierarchical Clustering provides better interpretability through dendrograms. Proper preprocessing and hyperparameter tuning are crucial for achieving meaningful clustering results.